INF1006: Computer Networks

TRIMESTER 2, AY 2025/26

Team Project: Network Application

1. Objectives

This project aims to enable students to apply computer network knowledge in practical network applications, offering hands-on experience in creating solutions for device communication. It presents real-world challenges in configuring network architecture, connecting devices, and ensuring seamless communication. Additionally, this project promotes learning about teamwork dynamics by exposing students to various potential issues related to effective collaboration.

2. Rules and Regulations

- This project requires teams of 4–6 students to design, implement, and test an application.
- The implementation must include 2-3 Raspberry Pis connected via wired or wireless LAN.
- Serial communication is not permitted.
- The Raspberry Pis must communicate with each other through the program developed by the team.
- The sensors or devices used in the project MUST be selected from the provided sensor/device list unless prior approval is obtained from the instructor.
- For the designated topics, a recommended set of sensors will be provided. Teams proposing their own project topic must clearly specify the required sensors/devices, selected from the provided sensor/device list.

3. Project Topics

Teams can choose to implement any (**not limited to**) of the following topics:

1. Environmental Monitoring
2. Smart Agriculture
3. Home Automation / Smart Lighting
4. Smart Metering
5. Smart Waste Bin

4. Deliverables & Submission

a. **Project proposal:** Teams are required to submit a one-page project proposal, which includes the students' names, student IDs, selected project title, abstract and expected goal, project plan (e.g., timeline, sensor/device information, and resources) and workload distribution. The proposal should be submitted to xSite Dropbox by the team leader by **Friday, 13th February 2026 23:59:59.**

b. **Final product and network diagram:** Teams are required to submit the final completed application, which must include your source code and any other files that are required to run the application. The team should also submit the network diagram drawn by using Cisco Packet.

c. **Poster and 3-minute video:** In the poster, the team should include the network diagram captured from Cisco Packet Tracer to illustrate how the devices are communicating with each other. The video should concentrate on the network configuration, how the application works and demonstrate all features of the application. It should contain contributions from all the team members. The team can choose to add a video blog too, but it is optional.

   The team leader must submit the network diagram, poster, the short video, and ONE zip file containing the complete application files to xSite Dropbox **by Thursday, 26th March 2026 23:59:59.**

5. Final Demonstration

A demo session will be arranged on **Week 12 (Friday, 27th March 2026)** during lab class. The team should first illustrate and explain the network diagram, followed by a demonstration of the network application functions. During the demonstration, the assessment team, comprising Faculty, PO members and TAs, will ask questions. Further details regarding the demonstration schedule will be provided at a later date.

6. Teamwork & Peer Assessment

The final mark for each team member is determined by both the team project mark (awarded to the entire team) and the peer assessment (<u>which may vary based on individual peer reviews within the team</u>). If all members contribute equally, each member will receive the same final mark as the project mark. However, if there are disparities in contributions, those who contribute less or not at all may receive lower scores, and in the worst case, no score. For instance, if Team X receives 25 marks for the project and team members A, B, C and D contribute equally, each receives 25 marks. However, if A contributes more and D contributes less (e.g., the peer assessment result is A: 110%, B and C: 100%, D: 90%), then the final mark is A: 27.5, B and C: 25, D: 22.5. Nonetheless, the maximum mark for the project remains 30.

Effective teams consist of members who contribute equally, leveraging their individual strengths and weaknesses. It is important for team members to negotiate and distribute the workload among themselves, considering each member's capabilities and expertise.

Each member of the team is asked to provide the peer assessment of each member of their team (including themselves), in terms of the quality of work and effort expended on the project and assignment. The online peer assessment system **TeamMates** (https://teammatesv4.appspot.com) will be used in this module.

7. Marking Scheme

This project weight 30% of the total assessment of this module. **Please refer to the detailed rubrics posted on xSiTe.**

Criteria	Marks
Project proposal: • The topic and abstract of the project. • Project goals, outcomes, and scope. • The project plan. • Sensors/devices to be used in this project.	5
Application demonstration and network diagram exposition: • Implementation of all proposed application features. • The Cisco Packet Tracer network diagram should clearly illustrate device communication topology.	15
Poster and Video: • Able to provide thorough overview and features of the use case. • Able to show how the networking applied in their application. • Easy to follow and understand.	10
Peer review	As described in part 6

Please note that project assessments will NOT factor in the use of advanced/expensive/new sensors or devices purchased for this project. Key guidelines:

- A pool of commonly used sensors is available for loans. Details about borrowing procedures and available equipment will be shared on xSite.
- Please ensure that only the available sensors are used. We do not recommend purchasing additional sensors or devices, as they cannot be claimed or reimbursed by this module.
- The list of sensors/devices to be used in the project MUST be included in the proposal.

8. Late Submission

A penalty of **20%** per day of this project marks (including Sunday and public holiday) will be imposed for late submission unless extension has been granted by the instructor **prior** to the submission date. Request for extension will be granted on a case-by-case basis. Any work submitted more than 4 days after the submission date will not be accepted and no mark will be awarded.

9. Notes on Plagiarism

The University's policy on copying does not allow you to copy software as well as your assessment solutions from another person. Copying of another person's work is unacceptable. It is the responsibility of all students that their assessment solutions are their own work. You must also ensure that others do not obtain access to your solutions for copying a part of them. Where such plagiarism is detected, both assessments involved will receive **ZERO** mark.

10. The Usage of AI Tools

- You may use AI programs e.g., ChatGPT to help generate ideas and brainstorm. However, you should note that the material generated by these programs may be inaccurate, incomplete, or otherwise problematic. Beware that use may also stifle your own independent thinking and creativity.
- You may not submit any work generated by an AI program as your own. If you include **material/code** generated by an AI program, it **MUST** be cited like any other reference material (with due consideration for the quality of the reference, which may be poor).
- Any plagiarism or other form of cheating will be dealt with severely under relevant SIT policies.
- SIT GenAI Guidelines can be found here: https://libguides.singaporetech.edu.sg/citation/aichatbot#s-lg-box-23111596